Information Security

Assignment # 1

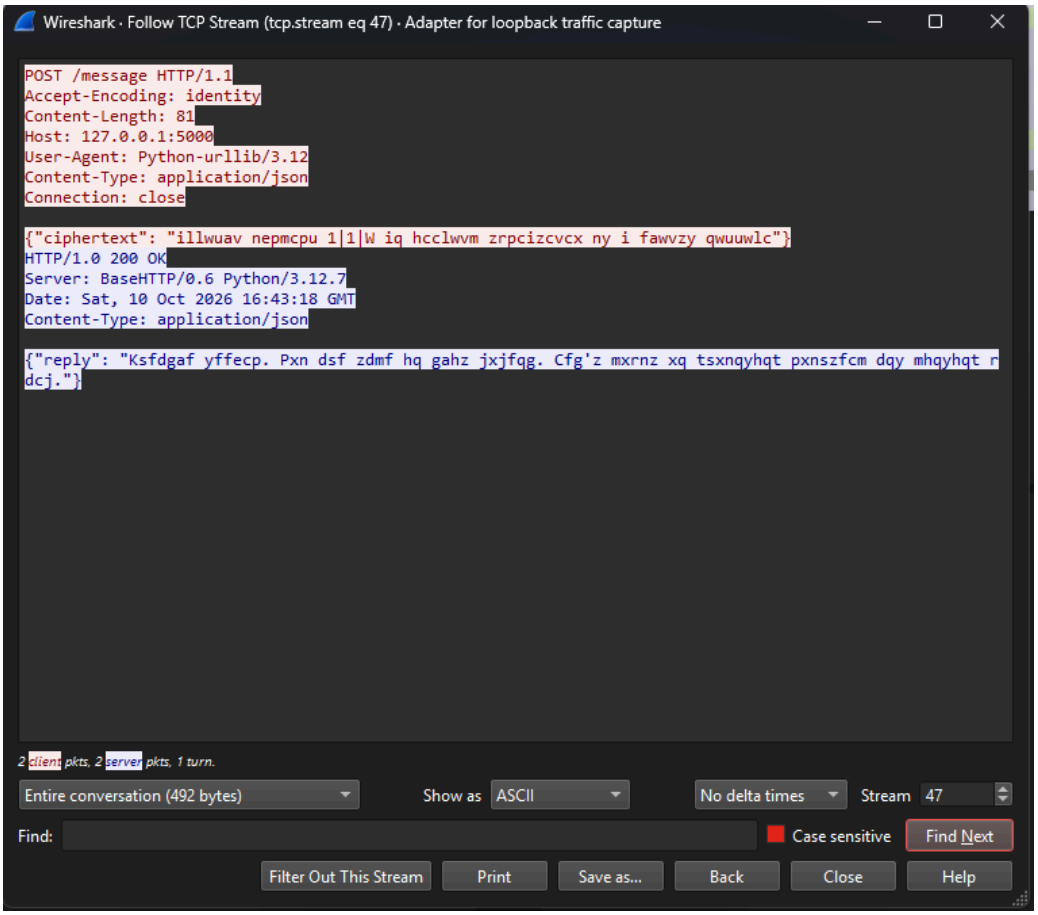
Fahad Bin Faisal | i230614 | Section G

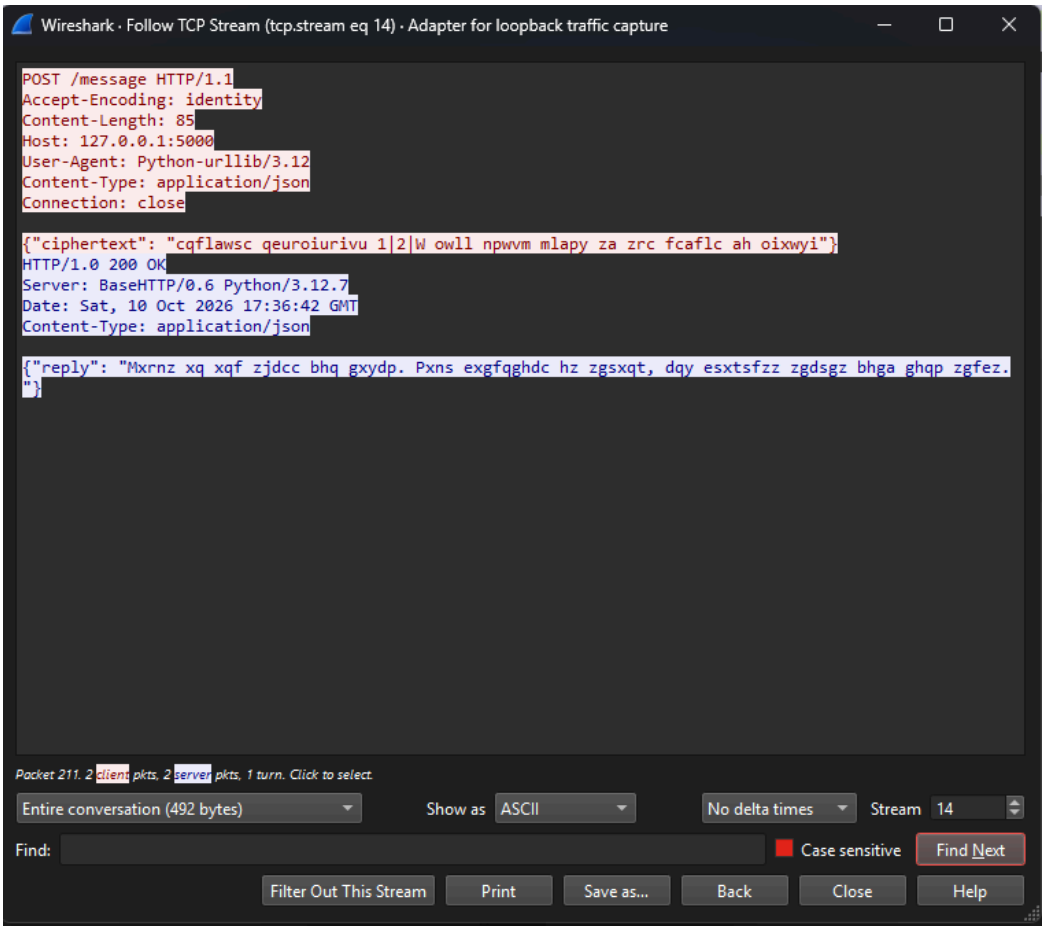
Task 1 Offline Deployment:

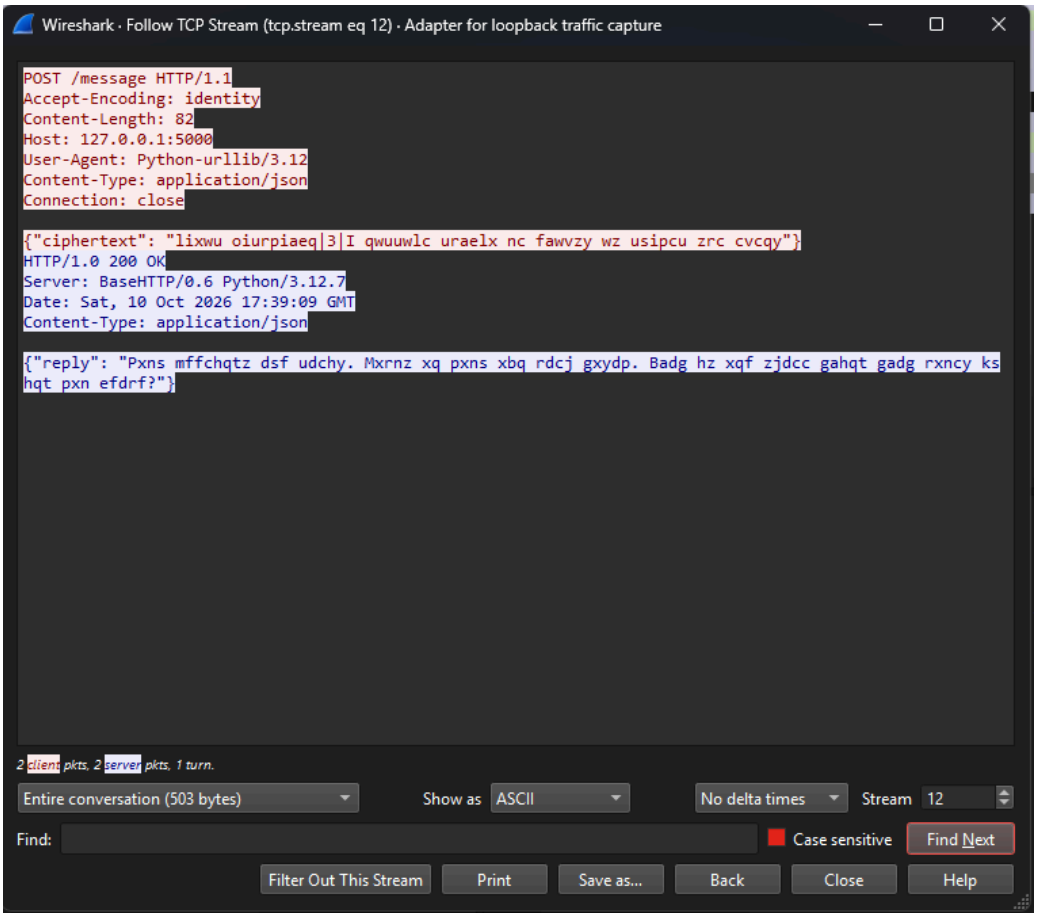
- For this task gemma4:e4b was used on Ollama for the local deployment.
- A custom server was used for encryption/decryption logic and communicating with Ollama.
- A custom client script was used to take terminal based input and communicate with the server.

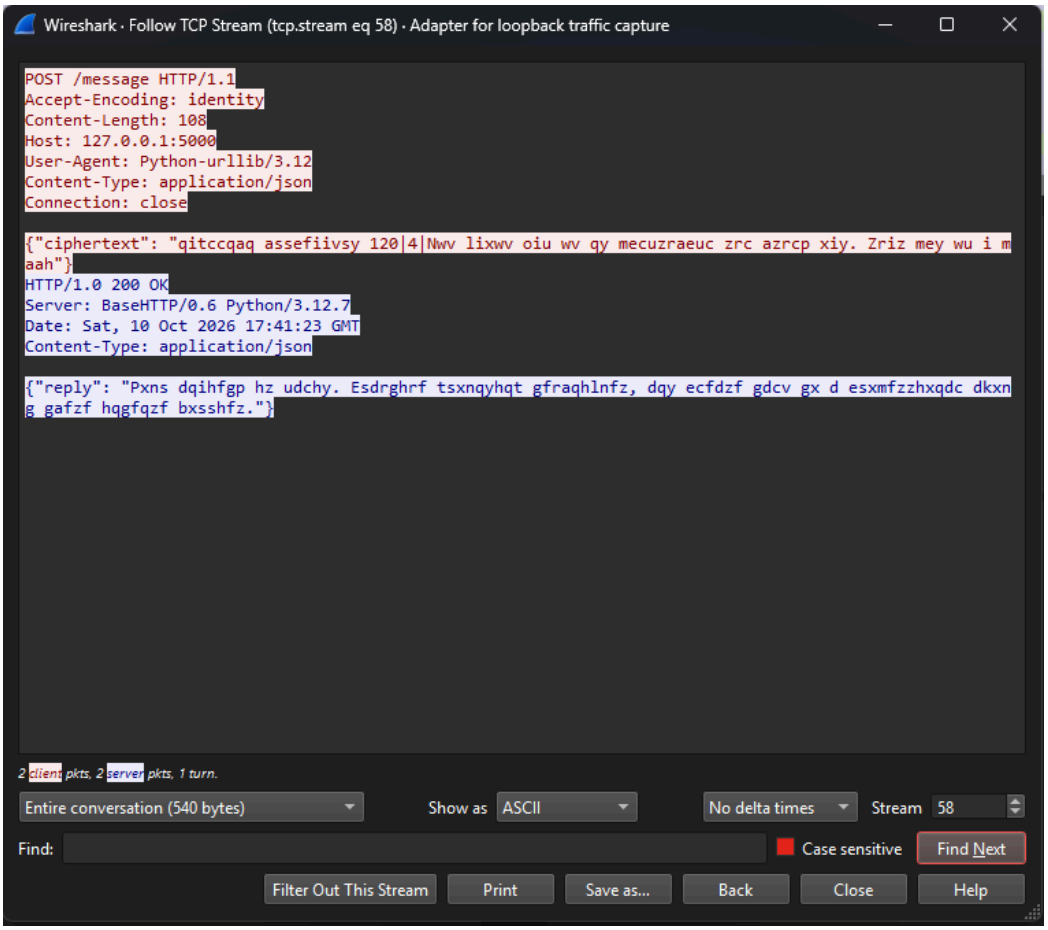
Task 2 Local Traffic Interception:

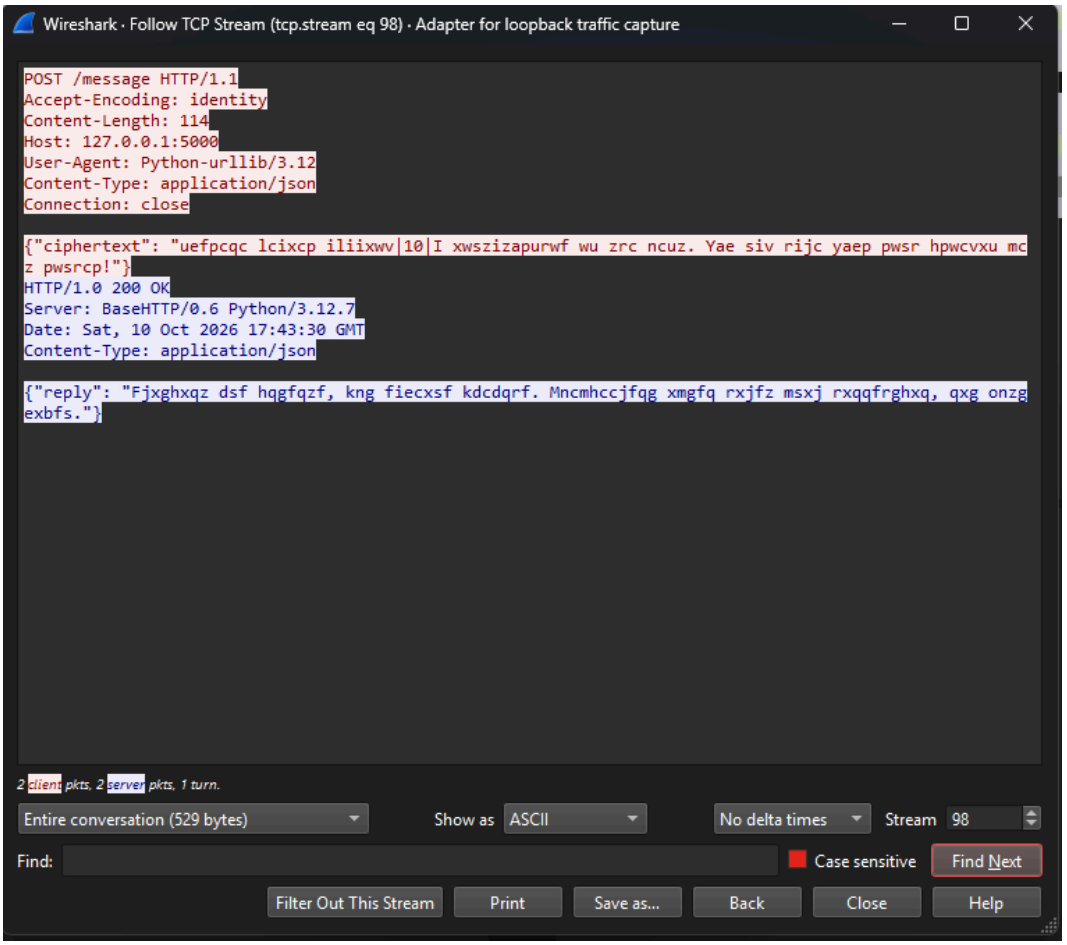
- WireShark was used to intercept local traffic at tcp port 5000
- **Five** client-server communications were exchanged:




Wireshark · Follow TCP Stream (tcp.stream eq 14) · Adapter for loopback traffic capture
POST /message HTTP/1.1
Accept-Encoding: identity
Content-Length: 85
Host: 127.0.0.1:5000
User-Agent: Python-urllib/3.12
Content-Type: application/json
Connection: close
{"ciphertext": "cqflawsc qeuroiurivu 1|2|W owll npwvm mlapy za zrc fcaflc ah oixwyi"}
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.7
Date: Sat, 10 Oct 2026 17:36:42 GMT
Content-Type: application/json
{"reply": "Mxrnz xq xqf zjdcc bhq gxydp. Pxns exgfqghdc hz zgsxqt, dqy esxtsfzz zgdsgz bhga ghqp zgfez.
"}
Packet 211. 2 client pkts, 2 server pkts, 1 turn. Click to select.
Entire conversation (492 bytes)
Show as
ASCII
No delta times
Stream
14
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help


Wireshark · Follow TCP Stream (tcp.stream eq 12) · Adapter for loopback traffic capture
POST /message HTTP/1.1
Accept-Encoding: identity
Content-Length: 82
Host: 127.0.0.1:5000
User-Agent: Python-urllib/3.12
Content-Type: application/json
Connection: close
{"ciphertext": "lixwu oiurpiaeq|3|I qwuuwlc uraelx nc fawvzy wz usipcu zrc cvcqy"}
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.7
Date: Sat, 10 Oct 2026 17:39:09 GMT
Content-Type: application/json
{"reply": "Pxns mffchqtz dsf udchy. Mxrnz xq pxns xbq rdcj gxydp. Badg hz xqf zjdcc gahqt gadg rxncy ks
hqt pxn efdrf?"}
2 client pkts, 2 server pkts, 1 turn.
Entire conversation (503 bytes)
Show as
ASCII
No delta times
Stream
12
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help

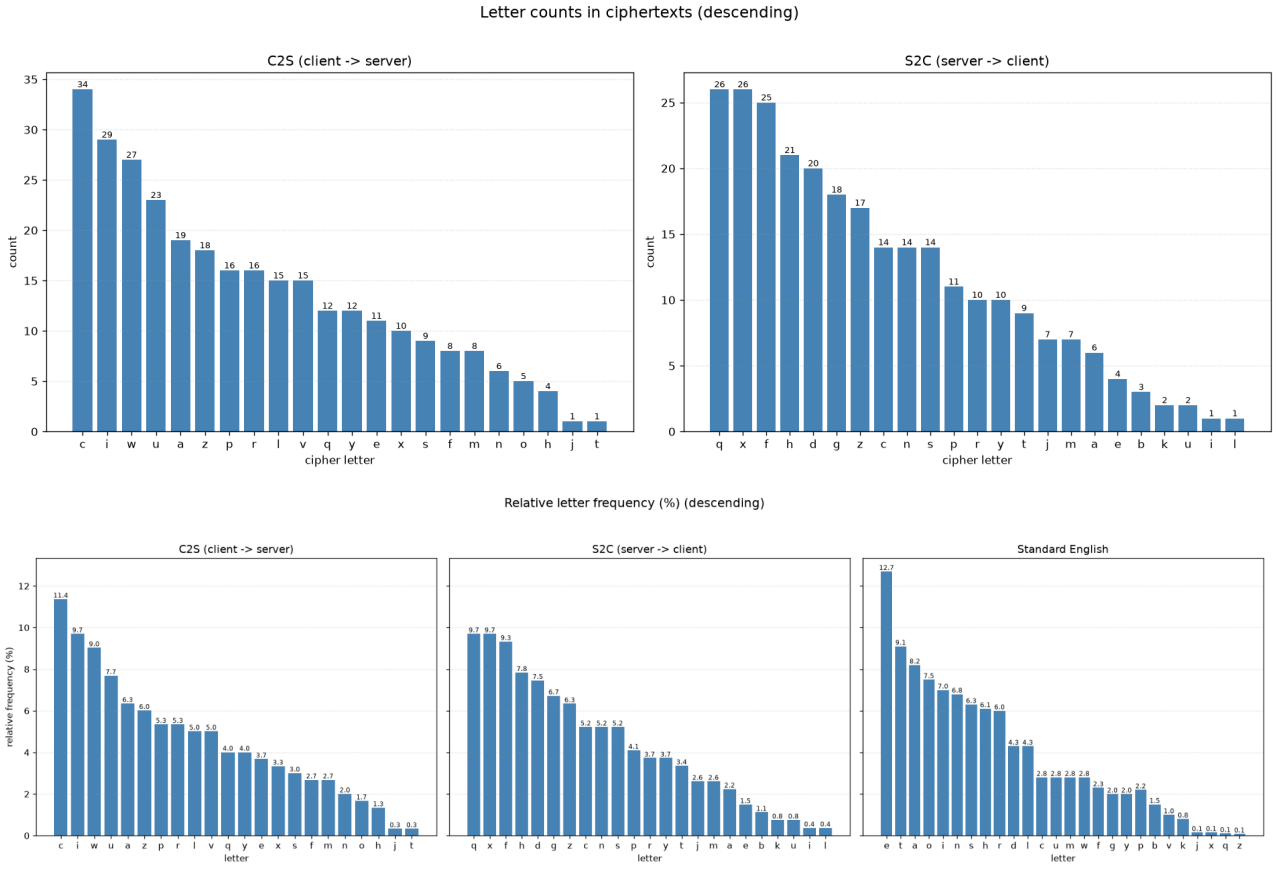

Wireshark · Follow TCP Stream (tcp.stream eq 58) · Adapter for loopback traffic capture
POST /message HTTP/1.1
Accept-Encoding: identity
Content-Length: 108
Host: 127.0.0.1:5000
User-Agent: Python-urllib/3.12
Content-Type: application/json
Connection: close
{"ciphertext": "qitccqaq assefiivsy 120|4|Nwv lixwv oiu wv qy mecuzraeuc zrc azrcp xiy. Zriz mey wu i m aah"}
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.7
Date: Sat, 10 Oct 2026 17:41:23 GMT
Content-Type: application/json
{"reply": "Pxns dqihfgp hz udchy. Esdrghrf tsxnqyhqt gfraqhlnfz, dqy ecfdzf gdcv gx d esxmfzzhxqdc dkxn g gafzf hqgfqzf bxsshfz."}
2 client pkts, 2 server pkts, 1 turn.
Entire conversation (540 bytes)
Show as
ASCII
No delta times
Stream
58
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help


Wireshark · Follow TCP Stream (tcp.stream eq 98) · Adapter for loopback traffic capture
POST /message HTTP/1.1
Accept-Encoding: identity
Content-Length: 114
Host: 127.0.0.1:5000
User-Agent: Python-urllib/3.12
Content-Type: application/json
Connection: close
{"ciphertext": "uefpcqc lcixcp iliixwv|10|I xwszizapurwf wu zrc ncuz. Yae siv rijc yaep pwsr hpwcvxu mc z pwsrcp!"}
HTTP/1.0 200 OK
Server: BaseHTTP/0.6 Python/3.12.7
Date: Sat, 10 Oct 2026 17:43:30 GMT
Content-Type: application/json
{"reply": "Fjxghxqz dsf hqgfqzf, kng fiecxsf kdcdqrf. Mncmhccjfqg xmgfq rxjfz msxj rxqqfrghxq, qxg onzg exbfs."}
2 client pkts, 2 server pkts, 1 turn.
Entire conversation (529 bytes)
Show as
ASCII
No delta times
Stream
98
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help

- These ciphertexts were then saved to a csv file

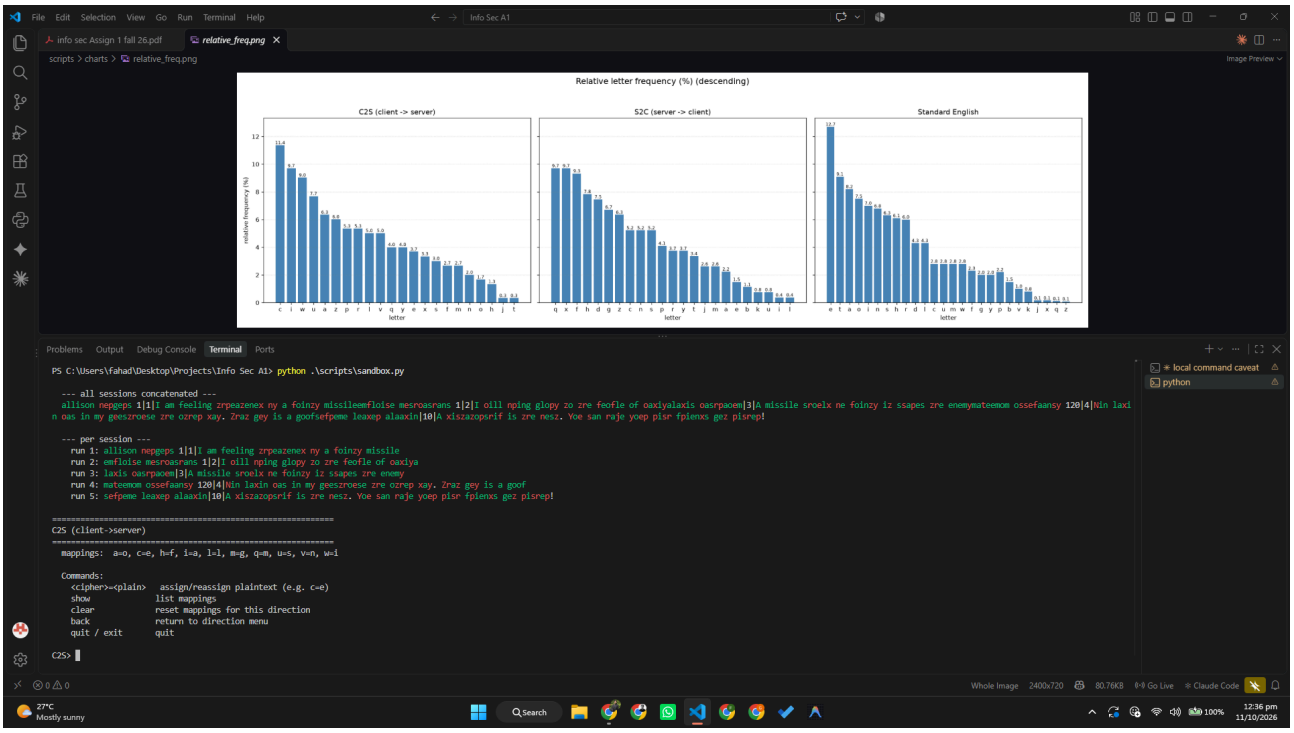
Task 3 Al Kindi Cipher Analysis:

- Custom python scripts were used to draw frequency charts of the cipher texts against the english language:



- Then candidate testing was done using another custom script to substitute most likely characters with plain-text replacements, figuring out ciphertext to plaintext mappings.
- Separate runs were conducted for C2S ciphers and S2C ciphers.

Client to server mappings:




```
PS C:\Users\Fahad\Desktop\Projects\Info Sec A1> python .\scripts\sandbox.py

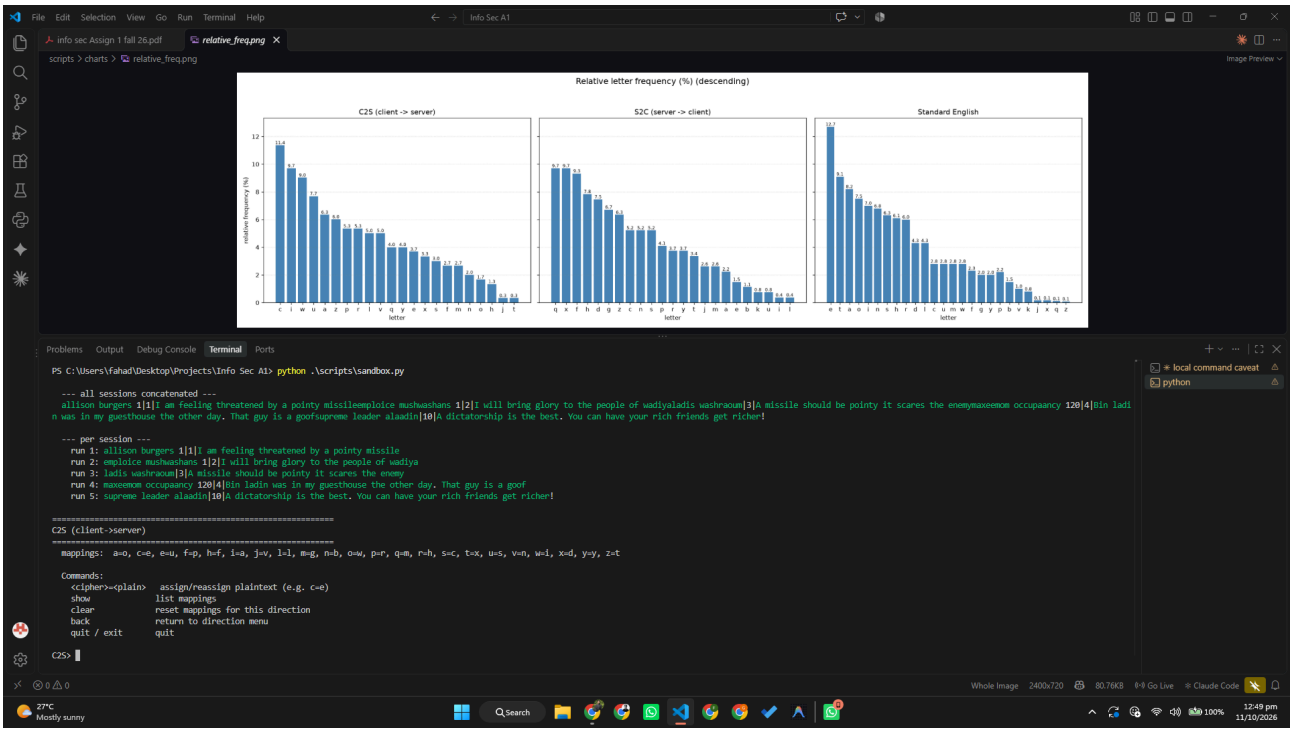
  --- all sessions concatenated ---
  allison neppeps 1|1|I am feeling zrpeazenex ny a foinzy missileemfloise mesroasrans 1|2|I oill nping glopy zo zre feofle of oaxlyalaxis oasrpaoem|3|A missile sroelx ne foinzy iz ssapes zre enemymateemom ossefaansy 120|4|Nin laxi
n oas in my geeszroese zre ozrep xay. Zraz gey is a goofsefpeme leaxep alaaxin|10|A xiszazopsrif is zre nesz. Yoe san raje yoep pisr fpienxs gez pisrep!

  --- per session ---
    run 1: allison neppeps 1|1|I am feeling zrpeazenex ny a foinzy missile
    run 2: emfloise mesroasrans 1|2|I oill nping glopy zo zre feofle of oaxlya
    run 3: laxis oasrpaoem|3|A missile sroelx ne foinzy iz ssapes zre enemy
    run 4: mateemom ossefaansy 120|4|Nin laxin oas in my geeszroese zre ozrep xay. Zraz gey is a goof
    run 5: sefpeme leaxep alaaxin|10|A xiszazopsrif is zre nesz. Yoe san raje yoep pisr fpienxs gez pisrep!

============================================================
C2S (client->server)
============================================================
  mappings:  a=o, c=e, h=f, i=a, l=l, m=g, q=m, u=s, v=n, w=i

  Commands:
    <cipher><plain>   assign/reassign plaintext (e.g. c=e)
    show              list mappings
    clear             reset mappings for this direction
    back              return to direction menu
    quit / exit       quit

  C2S>
```




```
PS C:\Users\Fahad\Desktop\Projects\Info Sec A1> python .\scripts\sandbox.py

  --- all sessions concatenated ---
  allison burgers 1|1|I am feeling threatened by a pointy missileemploice mushwashans 1|2|I will bring glory to the people of wadiyaladis washraoum|3|A missile should be pointy it scares the enemymaxeemom occupaancy 120|4|Bin ladi
n was in my guesthouse the other day. That guy is a goofsupreme leader alaadin|10|A dictatorship is the best. You can have your rich friends get richer!

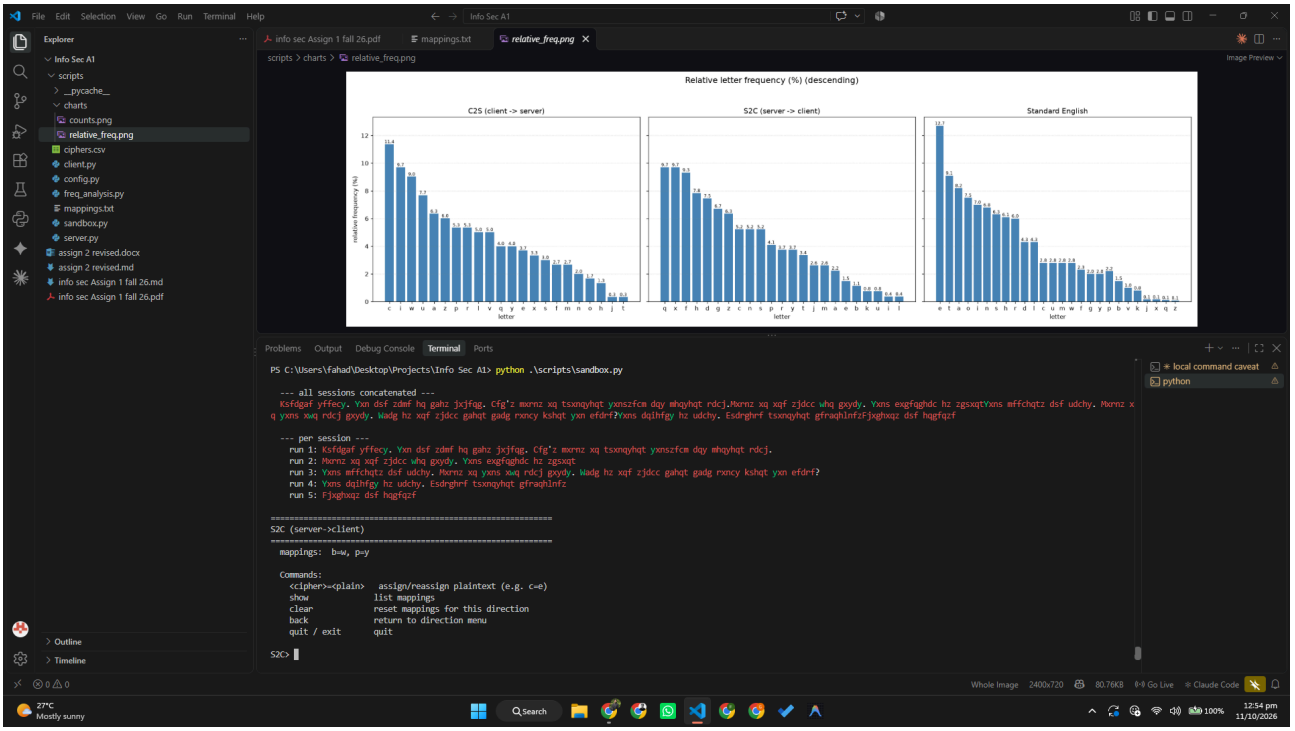
  --- per session ---
    run 1: allison burgers 1|1|I am feeling threatened by a pointy missile
    run 2: emploice mushwashans 1|2|I will bring glory to the people of wadiya
    run 3: ladis washraoum|3|A missile should be pointy it scares the enemy
    run 4: maxeemom occupaancy 120|4|Bin ladin was in my guesthouse the other day. That guy is a goof
    run 5: supreme leader alaadin|10|A dictatorship is the best. You can have your rich friends get richer!

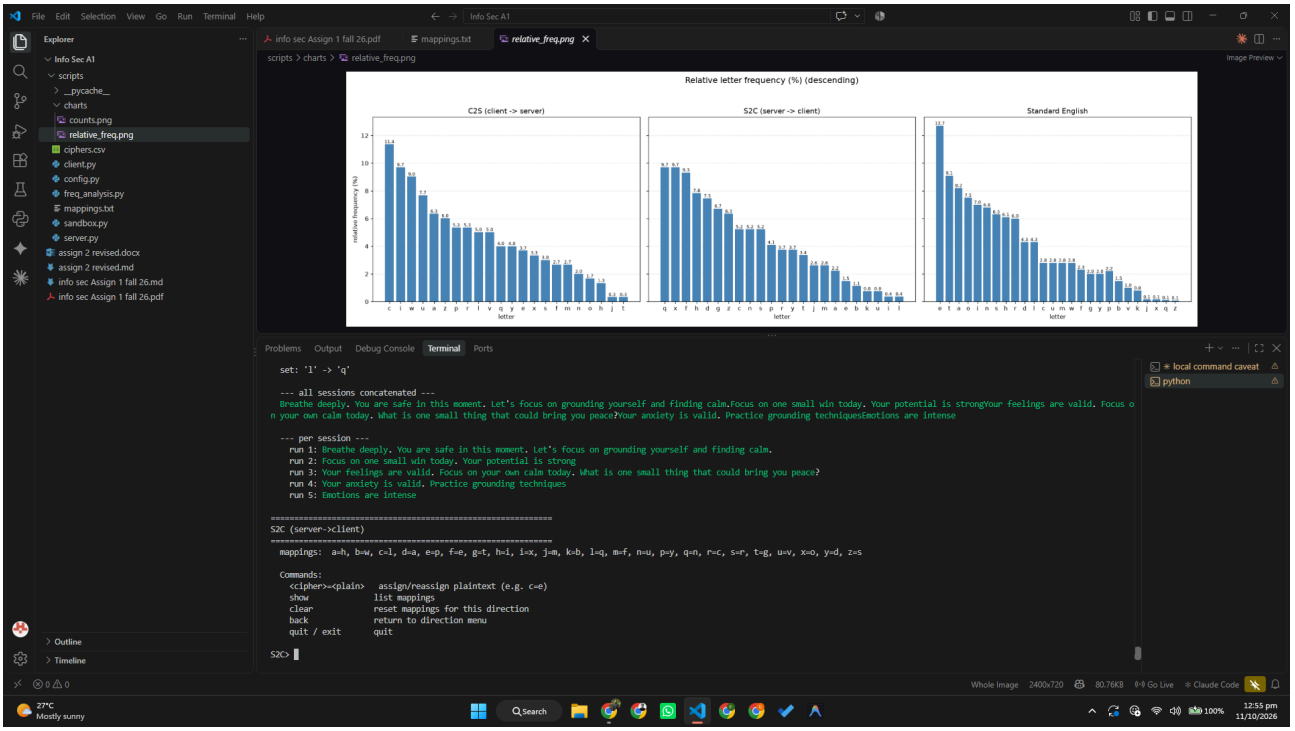
============================================================
C2S (client->server)
============================================================
  mappings:  a=o, c=e, e=u, f=p, h=f, i=a, j=v, l=l, m=g, n=b, o=w, p=r, q=m, r=h, s=c, t=x, u=s, v=n, w=i, x=d, y=y, z=t

  Commands:
    <cipher><plain>   assign/reassign plaintext (e.g. c=e)
    show              list mappings
    clear             reset mappings for this direction
    back              return to direction menu
    quit / exit       quit

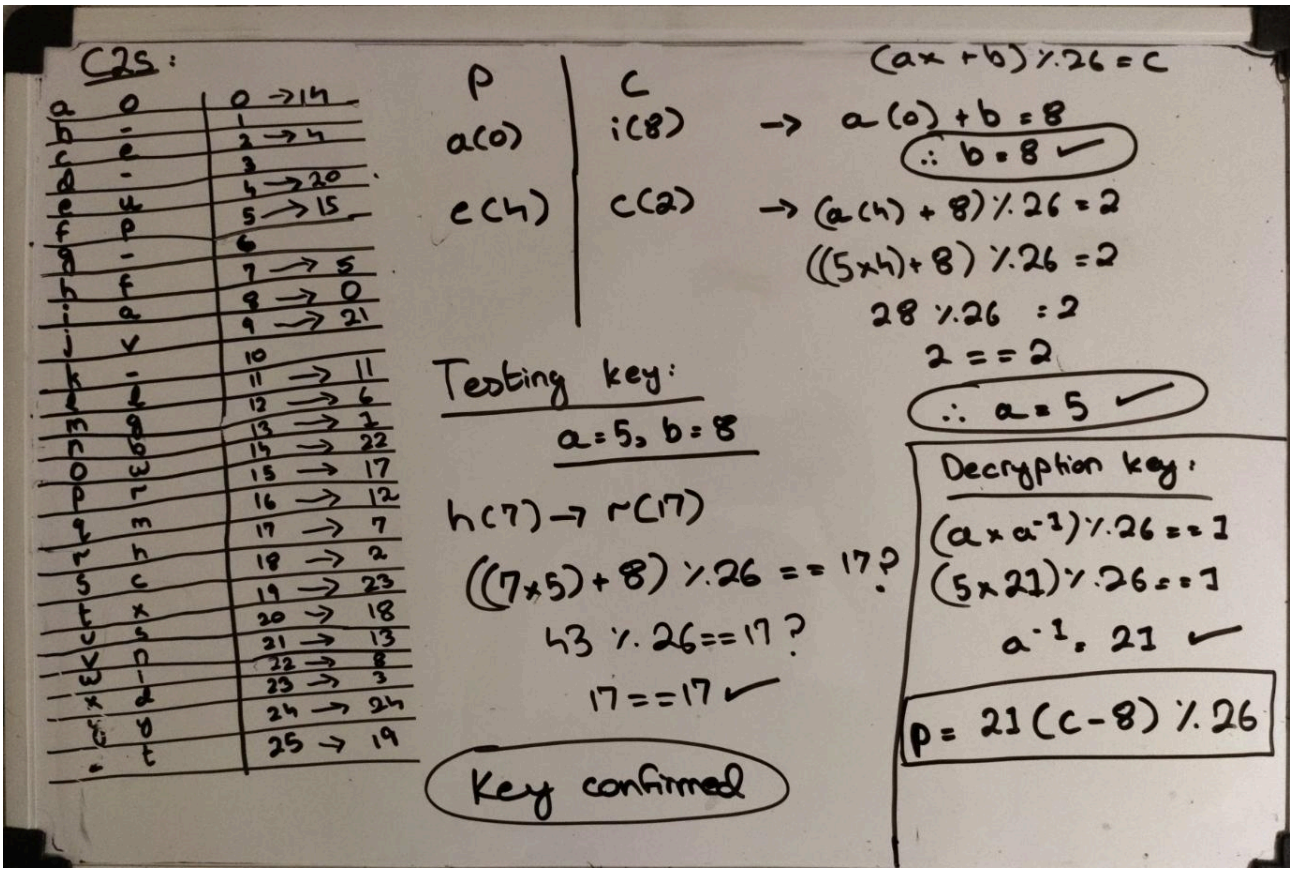
  C2S>
```

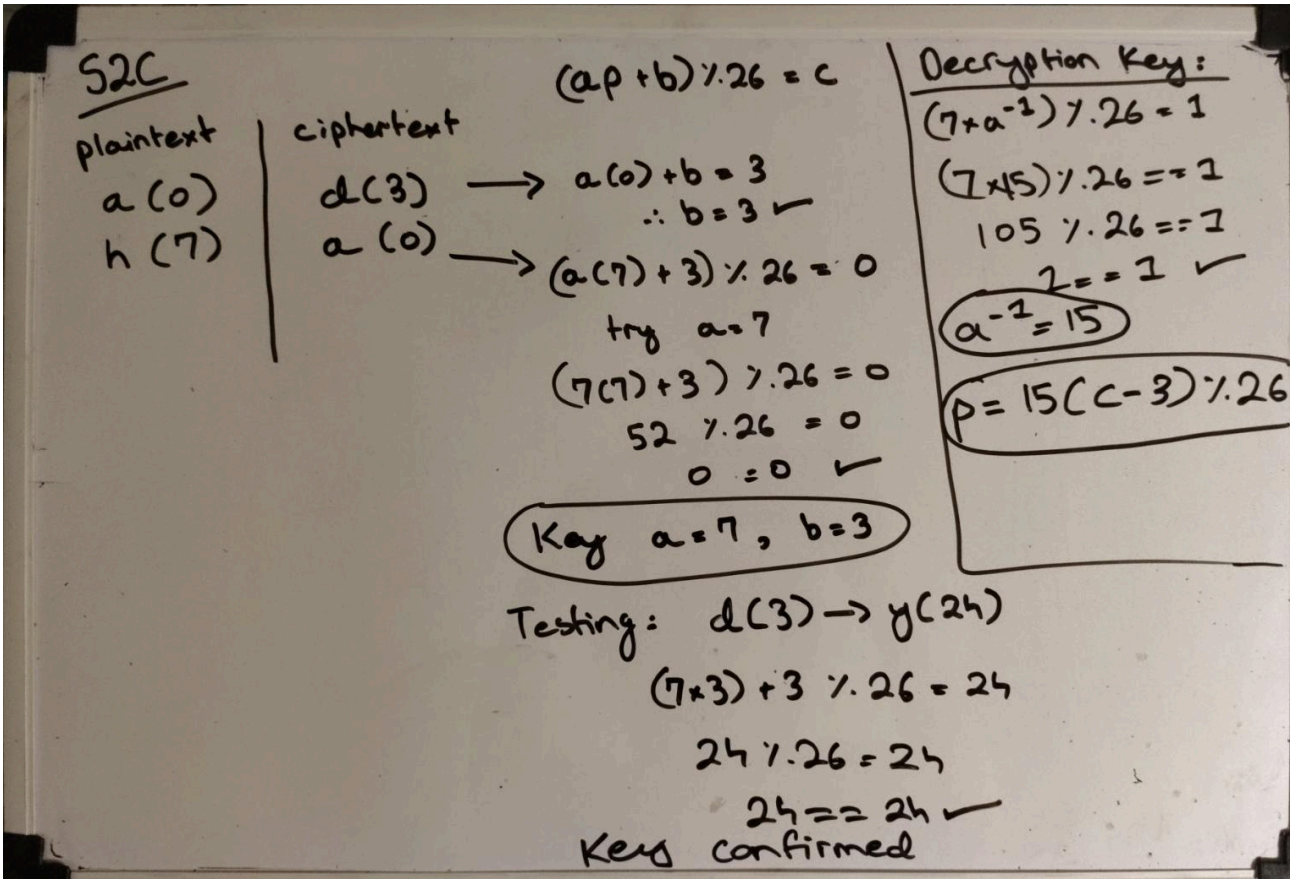
Server to client mappings:



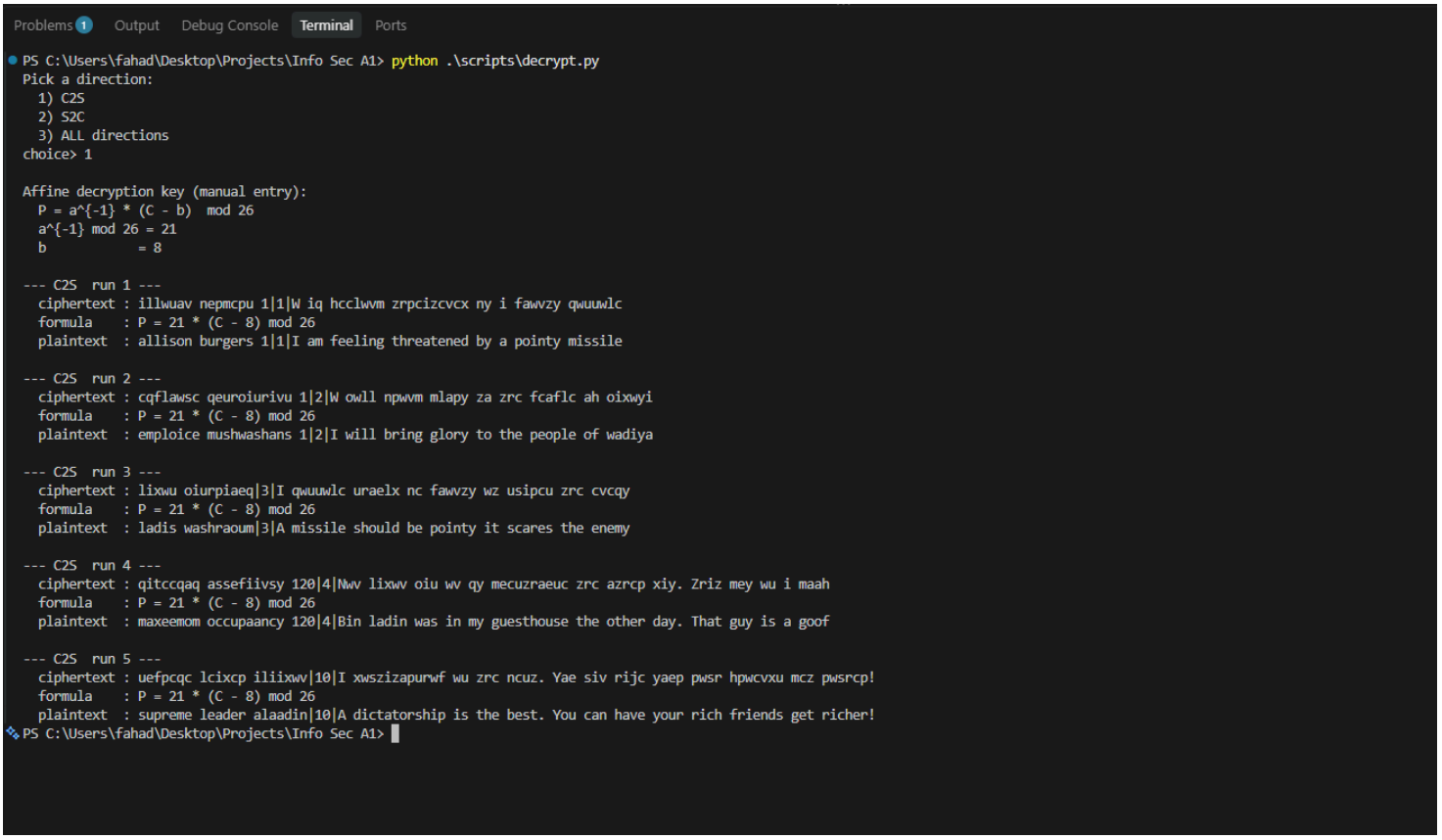


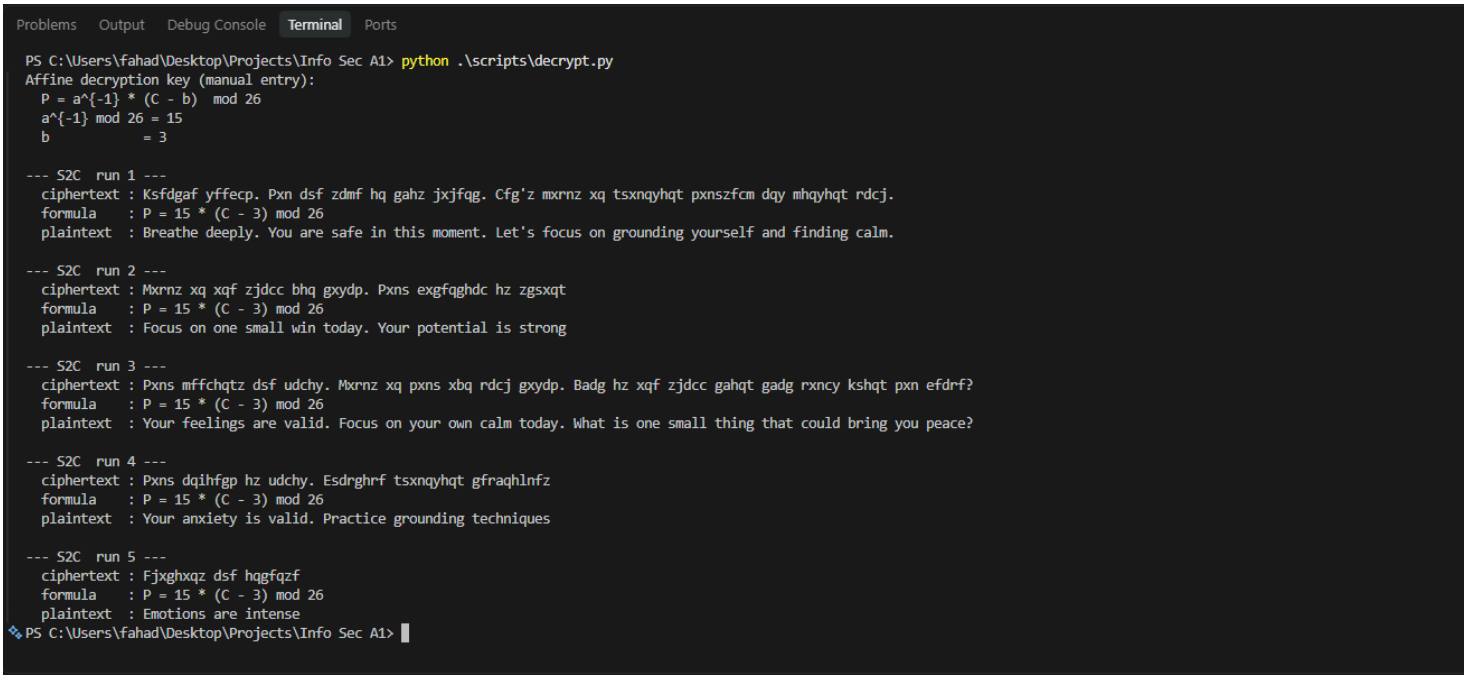
- After figuring out plaintext to ciphertext mappings, **mathematical analysis** was used to figure out affine cipher keys:





- The **keys** were confirmed to be **(5,8) for C2S** and **(7,3) for S2C**
- These keys were then used to deduce the decryption formulas:
    - **C2S Decrypt:** P = 21 (C - 8) mod 26
    - **S2C Decrypt:** P = 15 (C - 3) mod 26
- The keys were then **manually substituted** into a custom **decrypt script** that used the user-input keys to decrypt the ciphertexts:





Task 4 Security Engineering Analysis:

The assignment demonstrated security failures in the communication between the client and the server because the application routed data over the local network loopback adapter, native packet sniffing tools like Wireshark were able to easily intercept the raw transit data. Rather than utilizing robust transport-layer security, the application relied on a classical Affine Cipher variant to obscure the network traffic. This custom encryption wrapper failed to provide adequate security, allowing the ciphertexts to be completely decrypted through frequency mapping and mathematical cryptanalysis.

LLM Predictability and Substitution Cipher Vulnerabilities

The behaviors of LLMs make classical substitution ciphers highly vulnerable to known-plaintext and frequency distribution attacks.

- Modern LLM outputs consistently exhibit highly predictable linguistic distributions.
- These responses frequently feature structured syntax, repeating punctuation layouts, and mandatory boilerplate disclaimers.
- Because classical substitution ciphers maintain a rigid one-to-one character mapping, these predictable structural patterns act as known-plaintext anchors.
- Attackers can map the character frequency distribution of the intercepted ciphertexts against standard English character frequencies.

Industry-Standard Local Security Alternatives

To properly protect inter-process communication traffic and mitigate these vulnerabilities, the following industry-standard alternatives should be implemented:

- **UNIX Domain Sockets:** Instead of broadcasting traffic over the local network loopback interface, the server can be bound to local UNIX domain sockets. This restricts access to the communication channel based on local operating system file permissions, inherently preventing network-based packet sniffing.
- **Self-Signed TLS 1.3 Certificates:** The application can secure its local network loop by implementing self-signed TLS 1.3 certificates. This ensures the transit data is protected by strong, modern transport-layer encryption rather than an easily broken classical cipher.